Iteration and simulation

Warm-up question

Problem: 10 people are at a party, and all of them are wearing hats. They each place their hat in a pile; when they leave, they choose a hat at random. What is the probability at least one person selected the correct hat?

Question: Work with your neighbor to discuss the following question:

- Without calculating probabilities, how could you design an experiment to estimate this probability?

Designing an experiment

Step 1 : need 10 hats! (and 10 people)

(vector 1-10) Person 1 Person 2 ... Person 10

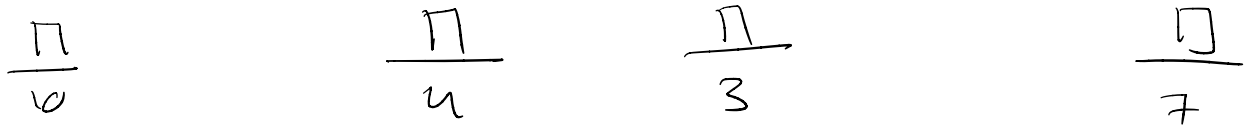
hat 1 hat 2 ... hat 10



Step 2 : randomly "shuffle" hats

(sampling in R) (randomly assign a hat to each person)

Person 1 Person 2 Person 3 ... Person 10



Step 3 : Who got their original hat?

(compare original #s to new ordering)

Step 4 : Repeat many times!

for loop!

Step 1: representing the hats

```r
hats <- 1:10

hats
```

```
 [1]  1  2  3  4  5  6  7  8  9 10
```

```r
hats[3]
```

```
[1] 3
```

- `hats` is a **vector**, containing the numbers 1 to 10
- entries in a vector are accessed by their index

Step 2: everyone draws a random hat

```
1 hats <- 1:10
2 randomized_hats <- sample(hats, size = 10, replace = FALSE)
3
4 hats
```

of items to sample

object to sample from

without replacement

```
[1]  1  2  3  4  5  6  7  8  9 10
```

```
1 randomized_hats
```

```
[1]  8  3  2  7  1  9  5  4 10  6
```

- The `sample` function creates a random sample from a vector
- How many people selected their original hat?

check if randomized_hats == hats (return T or F for each entry)

Step 3: check who got their original hat

```
hats <- 1:10
randomized_hats <- sample(hats, size = 10, replace = FALSE)
```

```
hats
```

```
[1]  1  2  3  4  5  6  7  8  9 10
```

```
randomized_hats
```

```
[1]  8  3  2  7  1  9  5  4 10  6
```

```
hats == randomized_hats
```

```
[1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
```

```
# TRUE is 1, FALSE is 0
sum(hats == randomized_hats)
```

← how many people got their original hat

```
[1] 0
```

```
# did at least one person get their hat?
sum(hats == randomized_hats) > 0
```

← did at least one person get their original hat?

```
[1] FALSE
```

Code so far

```
1 hats <- 1:10
2 randomized_hats <- sample(hats, size = 10, replace = FALSE)
```

```
1 sum(hats == randomized_hats) > 0
```

```
[1] FALSE
```

repeat these two lines many times

- Is this a good estimate of the probability?

No. Need to repeat

Step 4: iteration

A `for` loop repeats code many times:

```
1 nsim <- 10000 # number of simulations
2 for(i in 1:nsim){
3
4
5 }
```

"repeat the following code for i=1, i=2, i=3, ..., i= nsim"

Java:

```
for(int i= 0; i < n ; i++) {

:

}
```

Step 4: iteration

results : NA NA NA ... NA

(then fill this in as we repeat the code)

A `for` loop repeats code many times:

```
1  nsim <- 10000 # number of simulations
2  hats <- 1:10
3  results <- rep(NA, nsim) # vector to store results
4
5  for(i in 1:nsim){
6    randomized_hats <- sample(hats, size = 10, replace = FALSE)
7    results[i] <- sum(hats == randomized_hats) > 0
8  }
9
10 head(results)
```

"repeat NA nsim times"

```
[1] TRUE TRUE TRUE TRUE TRUE TRUE
```

i = 1 : (check)

TRUE NA NA NA ...

i = 2 : (check)

TRUE TRUE NA NA ... NA

i = 10000...

Step 4: iteration

A `for` loop repeats code many times:

```
nsim <- 10000 # number of simulations
hats <- 1:10
results <- rep(NA, nsim) # vector to store results

for(i in 1:nsim){
  randomized_hats <- sample(hats, size = 10, replace = FALSE)
  results[i] <- sum(hats == randomized_hats) > 0
}

mean(results)
```

or: `sum(results) / nsim`

```
[1] 0.6262
```

P(at least one person receives original hat) ≈ 0.626

- What if I wanted to repeat the simulation, with a different number of people?

Removing magic numbers

Without magic numbers:

```
 1 nsim <- 10000 # number of simulations
 2 n_people <- 10 # number of people
 3 hats <- 1:n_people
 4 results <- rep(NA, nsim) # vector to store results
 5
 6 for(i in 1:nsim){
 7   randomized_hats <- sample(hats,
 8                             size = n_people,
 9                             replace = FALSE)
10   results[i] <- sum(hats ==
11                       randomized_hats) > 0
12 }
13
14 mean(results)
```

```
[1] 0.6344
```

- Why did I get different results?

Final code

each time I run a process w/ "random" functions, I get same result

```
1  set.seed(3) # set a seed for reproducibility
2
3  n_people <- 10 # number of people at the party
4  hats <- 1:n_people # numbered hats
5  nsim <- 10000 # number of simulations
6  results <- rep(NA, nsim) # vector to store the results
7
8  for(i in 1:nsim){
9    # hats are randomly assigned to each person
10   randomized_hats <- sample(hats, n_people, replace = F)
11
12   # did at least one person get their hat back?
13   results[i] <- sum(randomized_hats == hats) > 0
14 }
15
16 mean(results)
```

Class activity

- Work with a neighbor on the class activity
- At the end of class, submit your work as an HTML file on Canvas (one per group, list all your names)